FINITE TYPE OF A MORPHISM OF FORMAL SPACES IS FINITE TYPE OF THE REDUCTIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof derives the (1)-(2)-(3) equivalence from the property-transfer lemma, computes the reduction of the affine pieces as the quotient by the ideals of topologically nilpotent elements, and closes via the etale-local nature of local finite type

Editorial note. The label lemma-finite-type-red-morphisms names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the ring-theoretic property of arrows of WAdm^count defined in lemma-finite-type-red of the same chapter
- the correspondence between properties of arrows of WAdm^count and properties of morphisms of formal algebraic spaces (Formal Spaces, Lemma formal-spaces-lemma-property-defines-property-morphisms)
- the reduction functor on formal algebraic spaces and its etale-local behaviour (Formal Spaces, Lemmas formal-spaces-lemma-reduction-formal-algebraic-space and formal-spaces-lemma-reduction-smooth, Example formal-spaces-example-reduction-affine-formal-spectrum)
- local finite type of morphisms of algebraic spaces is etale local on source and target (Morphisms of Spaces, Section spaces-morphisms-section-finite-type)

Source reference. Original definition 0AIM: sources/stacks-chow/tags.txt, source lines 13270–13270. Prerequisite pointer invoked at proof line(s) 2813; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0GB6: sources/stacks-chow/tags.txt, source lines 20361–20361. Prerequisite pointer invoked at proof line(s) 2854; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0ANG: sources/stacks-chow/tags.txt, source lines 13439–13439. Prerequisite pointer invoked at proof line(s) 2837; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AIN: sources/stacks-chow/tags.txt, source lines 13271–13271. Prerequisite pointer invoked at proof line(s) 2844; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0GB7: sources/stacks-chow/tags.txt, source lines 20362–20362. Prerequisite pointer invoked at proof line(s) 2846; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GC2: sources/stacks-restricted/pinned-original-source.tex, source lines 2631–2631. Prerequisite pointer invoked at proof line(s) 2810; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 03XE: sources/stacks-chow/tags.txt, source lines 5145–5145. Prerequisite pointer invoked at proof line(s) 2862; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of locally countably indexed formal algebraic spaces over S. The following are equivalent*

(1) *for every commutative diagram*

$$\begin{array}{ccc} U & \longrightarrow & V \\ \downarrow & & \downarrow \\ X & \longrightarrow & Y \end{array}$$

with U and V affine formal algebraic spaces, $U \to X$ and $V \to Y$ representable by algebraic spaces and étale, the morphism $U \to V$ corresponds to an arrow of $WAdm^{count}$ satisfying the property defined in Lemma 0GC2,

(2) *there exists a covering $\{Y_j \to Y\}$ as in Formal Spaces, Definition 0AIM and for each j a covering $\{X_{ji} \to Y_j \times_Y X\}$ as in Formal Spaces, Definition 0AIM such that each $X_{ji} \to Y_j$ corresponds to an arrow of $WAdm^{count}$ satisfying the property defined in Lemma 0GC2,*

(3) *there exist a covering $\{X_i \to X\}$ as in Formal Spaces, Definition 0AIM and for each i a factorization $X_i \to Y_i \to Y$ where Y_i is an affine formal algebraic space, $Y_i \to Y$ is representable by algebraic spaces and étale, and $X_i \to Y_i$ corresponds to an arrow of $WAdm^{count}$ satisfying the property defined in Lemma 0GC2, and*

(4) *the morphism $f_{red} : X_{red} \to Y_{red}$ is locally of finite type.*

Proof. The equivalence of (1), (2), and (3) follows from Lemma 0GC2 and an application of Formal Spaces, Lemma 0ANG. Let Y_j and X_{ji} be as in (2). Then

- The families $\{Y_{j,red} \to Y_{red}\}$ and $\{X_{ji,red} \to X_{red}\}$ are étale coverings by affine schemes. This follows from the discussion in the proof of Formal Spaces, Lemma 0AIN or directly from Formal Spaces, Lemma 0GB7.
- If $X_{ji} \to Y_j$ corresponds to the morphism $B_j \to A_{ji}$ of $WAdm^{count}$, then $X_{ji,red} \to Y_{j,red}$ corresponds to the ring map $B_j/\mathfrak{b}_j \to A_{ji}/\mathfrak{a}_{ji}$ where $\mathfrak{b}_j$ and $\mathfrak{a}_{ji}$ are the ideals of topologically nilpotent elements. This follows from Formal Spaces, Example 0GB6. Hence $X_{ji,red} \to Y_{j,red}$ is locally of finite type if and only if $B_j \to A_{ji}$ satisfies the property defined in Lemma 0GC2.

The equivalence of (2) and (4) follows from these remarks because being locally of finite type is a property of morphisms of algebraic spaces which is étale local on source and target, see discussion in Morphisms of Spaces, Section 03XE. □